Jason Sacerio

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Education

University of Central Florida — Orlando, FL
Bachelor of Science in Computer Science, Minor in Math — *Spring 2028*

Experience

Treasurer — Jan. 2026 - Present
Knight Hacks — *Orlando, FL*

- Led a **100-person team** as **Treasurer of Knight Hacks**, UCF's Largest student-run tech organization serving a community of **1500+ students**. Won **best event** at UCF RSO Ceremony 2026 and was nominated for **11** other awards
- Managed finances for a 501(c)(3) nonprofit, overseeing a **$100k+** annual budget across sponsorships and operating expenses. Secured **$90K+** in sponsorships from **Fortune 500** companies to support Knight Hacks IX

Researcher — Sept. 2025 - Present
UCF — *Orlando, FL*

- Conducted research on security and fault tolerance of **LLMs for embedded systems**, evaluating **15+ attack scenarios** across **6 hardware platforms** using real hardware and Gem5 simulation
- Analyzed **10+ mitigation strategies** for LLM vulnerabilities, including memory bit-flip attacks and **8 novel embedded-specific attack vectors**, across **6 embedded boards and mobile devices**

Software Engineer Intern — Apr. 2024 - Sept. 2024
Blackstone — *Miami, FL*

- Re-architected financial data processing pipelines using Pandas and NumPy vectorization, eliminating nested loops and reducing compute time by **40%** on **10M+ row datasets**.
- Optimized SQL queries, indexing strategies, and backend ETL workflows in Python, reducing end of day reporting latency by **30%** and improving throughput, maintainability, and reliability across financial ingestion pipelines.

Software Engineer Intern — Jan. 2023 – Aug. 2023
Capitol Steel Structures — *Miami, FL*

- Solely designed a comprehensive customer database using **MongoDB and Django**, supporting **1200+** client records and streamlining efficiency
- Built the company's first e-commerce website using **Python, HTML/CSS, and Stripe** for payment processing, enabling online sales of steel products and generating **$273,000** in revenue in the first year.

Projects

ZPM-TUNA | *C, C++, Python, ROS2, OpenCV, Arduino,*

- Won **1st place** in **Pheratech Systems ROS2 Challenge** during Knight Hacks VIII (2025)
- Developed **ROS2 based autonomous swarm drones** for evacuation scenarios, enabling coordinated navigation through simulated fire environments to locate optimal real-time exits
- Implemented inter-drone communication and adaptive pathfinding, achieving average route convergence to optimal exit in **sub 12 seconds** and reducing redundant path traversal by **27%** compared to baseline in simulations.

Conduit | *Python, pyTorch, OpenCV, Flask, Electron*

- Won **2nd place** in the **Healthcare Track at Hacklytics 2026 @ Georgia Tech**
- Built multimodal assistive HCI system integrating eye tracking with 3D iris projection, real-time ASL CV processing, achieving **sub 120ms** interaction latency
- Achieved **93%** gaze tracking stability after adaptive smoothing and **88%** EEG intent classification accuracy using engineered feature extraction and majority vote stabilization.

Guido | *C, C++, Python, ROS2, NAV2, NVIDIA SDK*

- Won **Most Innovative Hack** during **HackUSF 2026**
- Developed a fully autonomous wheelchair navigation system using ROS2, integrating LiDAR-based SLAM for real-time mapping. achieved **95%** obstacle avoidance success rate and **sub-100ms** planning latency.
- Architected a modular ROS2 autonomy stack with custom frontier based exploration, clustering and ranking frontiers by path cost and information gain. Enabling **40%** faster map coverage in unknown environments.

Technical Skills

Languages: C, C++, Rust, Python, Java, Javascript, Typescript, SQL, HTML/CSS
Libraries/Frameworks: Django, Tailwind, MongoDB, Pandas, NumPy, Matplotlib, ROS2
Developer Tools: Git, Docker, Jupyter Notebook, Visual Studio, PyCharm, CLion, Eclipse, Kubernetes, IDA Pro